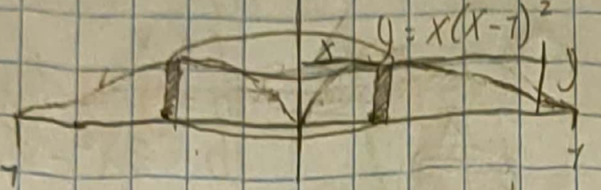
Tarea Sección 6.3

1)



$r = x$

$h = x(x-1)^2$

$C = 2\pi x$

$$V = 2\pi\int_0^1 x\left(x(x-1)^2\right)dx = x\left(x(x^2-2x+1)\right)dx$$

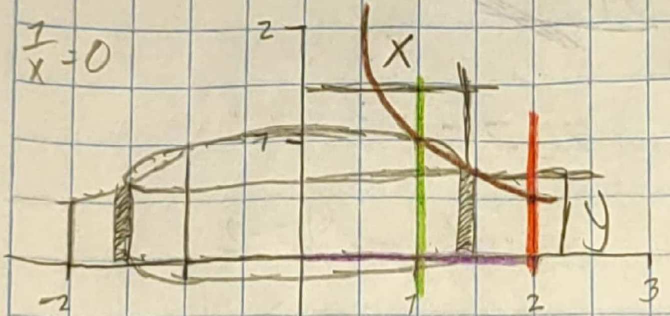
$$\int^1 x(x^3-2x^2+x)\,dx$$

$$V = 2\pi\int_0^1 x^4 - 2x^3 + x^2\,dx$$

$$V = 2\pi\left(\frac{x^5}{5} - \frac{1}{2}x^4 + \frac{x^3}{3}\right)\Big|_0^1 = \frac{\pi}{15}u^3$$

3) $y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 2$ eje y

$\frac{1}{x} = 0$



$r = x$ $h = y = \frac{1}{x}$

$$V = 2\pi\int_1^2 x\left(\frac{1}{x}\right)dx$$

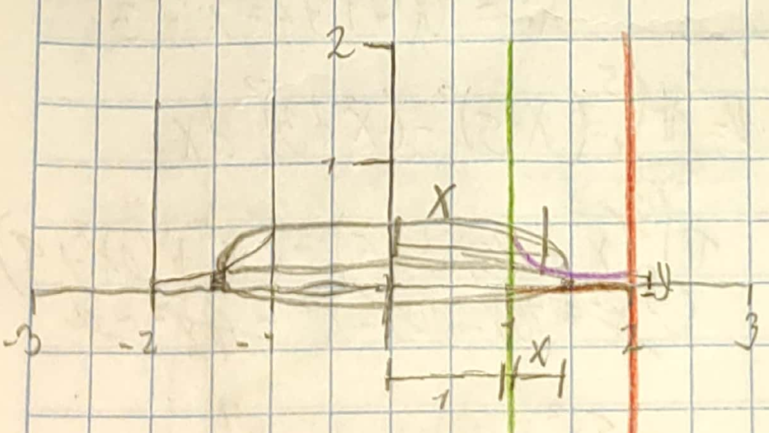
$$V = 2\pi\int_1^2 1\,dx = x\Big|_1^2 = 2-1$$

$$V = 2\pi\cdot 1 = 2\pi\,u^3$$

5) $y = e^{-x^2}$, $y = 0$, $x = 1$, $x = 2$

$2\pi r h\,dx$

$r = x$ $h = y = e^{-x^2}$



$$V = 2\pi\int_0^1 x\,e^{-x^2}dx = 2\pi\int_1^2 x e^m\cdot\frac{dm}{-2x}$$

$$V = 2\pi\left(-\frac{1}{2}\int_1^2 e^m\,dm\right)$$

$$V = 2\pi\left[-\frac{1}{2}\left(e^{x^{-2}}\right)\right]\Big|_{-1}^{0}$$

$$V = \pi\left(1 - \frac{1}{e}\right)$$

$m = -x^2$

$dm = -2x\,dx$

$\frac{dm}{2x} = dx$

7) $y_1 = x^2$, $y_2 = 6x - 2x^2$

dat.

$y_1 = y_2$

$x^2 = 6x - 2x^2$

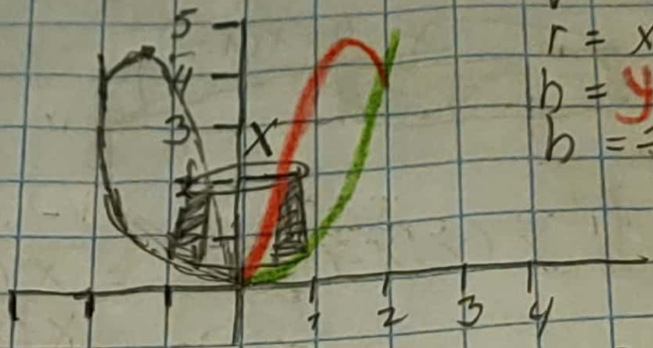
$6x - 2x^2 - x^2 = 0$

$6x - 3x^2 = 0$

$x(x-2) = 0$

$x = 0 , y = 0$

$x = 2 , y = 4$



$V = 2\pi \, r h \, dr$

$r = x$

$h = y - y = (6x - 2x^2) - x^2$

$h = -3x^2 - 6x$

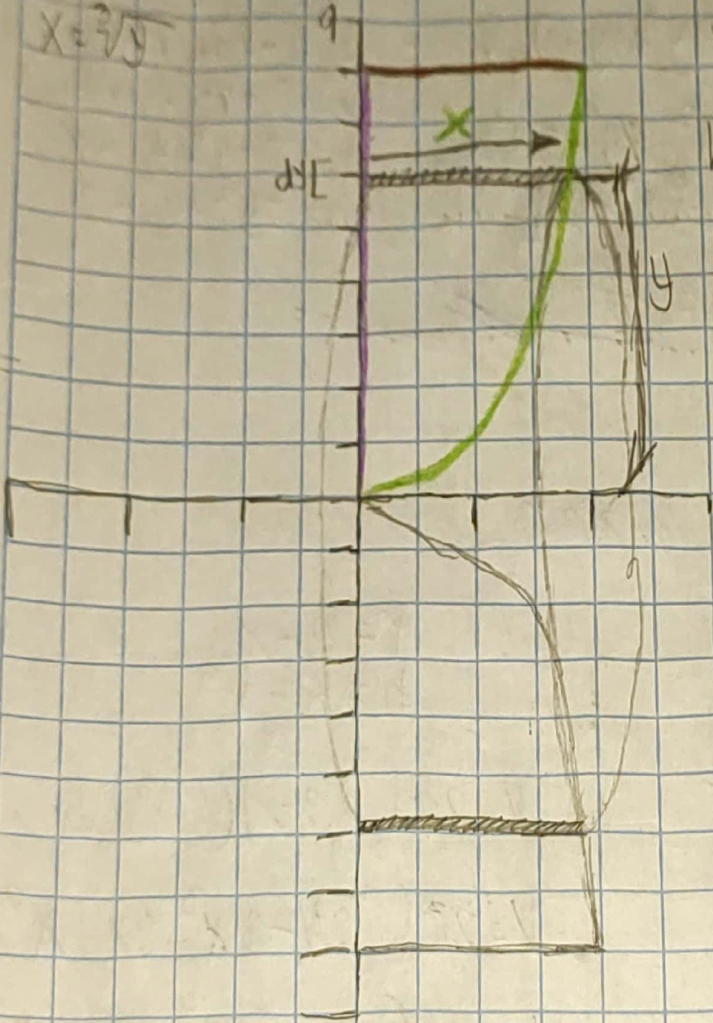
$V = 2\pi \int_0^2 x(3x^2 - 6x)\,dx = 2\pi \int_0^2 -3x^3 - 6x^2\,dx$

$V = 2\pi \left[-\frac{3}{4}x^4 - 2x^3 \right]\Big|_0^2$

$V = -(-8\pi) = \underline{8\pi}$

9) $y = x^3$, $y = 8$, $x = 0$

$x = \sqrt[3]{y}$



$V = 2\pi\, r\, h\, dy$ $\qquad r = y$

$h = x = \sqrt[3]{y}$

$V = \int_0^8 2\pi\, y\, \sqrt[3]{y}\, dy$

$V = 2\pi \int_0^8 y \cdot y^{1/3}\, dy = 2\pi \int_0^8 y^{4/3}\, dy$

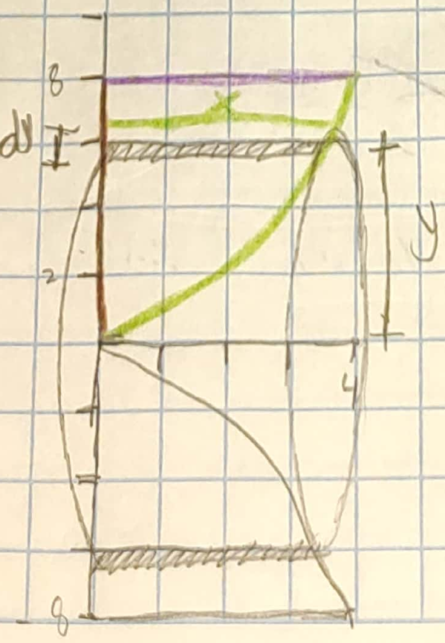
$V = 2\pi \left(\frac{3}{7} y^{7/3}\right)\Big|_0^8 = 2\pi\left(\frac{3}{7}(8)^{7/3}\right)$

$V = \frac{768}{7}\pi\ u^3$

11) $y = x^{3/2}$, $y = 8$, $x = 0$

$y = x^{3/2}$

$x = \sqrt[3]{y^2}$



$V = 2\pi\, r\, h\, dy$ $\qquad r = y$

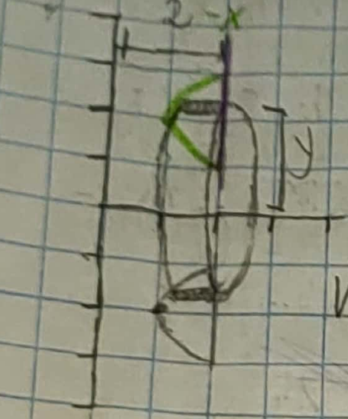
$h = x = \sqrt[3]{y^2}$

$V = 2\pi \int_0^8 y\, \sqrt[3]{y^2}\, dy$ $\quad y^{2/3}$

$V = 2\pi \int_0^8 y^{5/3} = 2\pi\left(\frac{3}{8} y^{8/3}\right)\Big|_0^8$

$V = 192\pi$

73) $x = 1+(y-2)^2$ $\qquad x = 2$ $\qquad V = 2\pi r\, h\, dy$



$r = y$

$h = 2 - x = 2-(1+(y-2)^2)$

$h = 2-(1+(y^2-4y+4))$

$h = 2-1-(y^2-4y+4)$

$h = 1-y^2+4y-4$

$h = -y^2+4y-3$

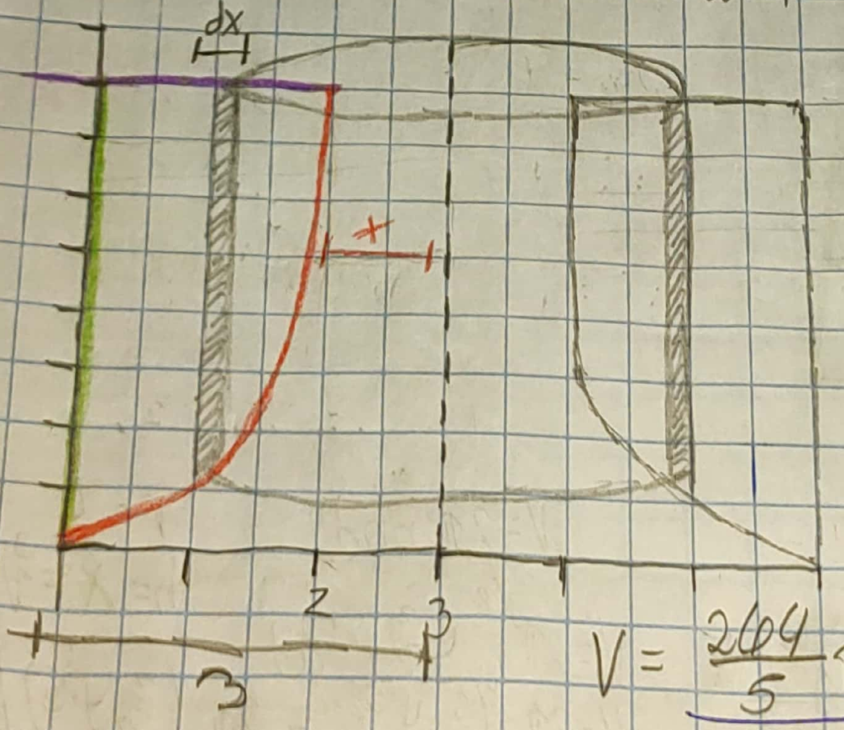
$V = 2\pi\int_1^3 y(-y^2+4y-3)\,dy$

$V = 2\pi\int_1^3 -y^3+4y^2-3y\,dy$

$V = 2\pi\left(-\frac{y^4}{4}+\frac{4}{3}y^3-\frac{3y^2}{2}\right)\Big|_1^3$

$V = \frac{9}{2}\pi - \left(-\frac{5}{6}\pi\right) = \frac{16}{3}\pi$

75) $y = x^3,\ y = 8,\ x = 0$ $\qquad$ alrededor $x = 3$



$r = 3-x$

$h = 8-x^3$

$V = 2\pi\int_0^2 (3-x)(8-x^3)\,dx$

$V = 2\pi\int_0^2 24-3x^3-8x+x^4\,dx$

$V = 2\pi\left(24x-\frac{3}{4}x^4-4x^2+\frac{x^5}{5}\right)\Big|$

$V = \frac{264}{5}\pi$

17) $y_1 = x^2$, $y_2 = 2 - x^2$ alrededor $x = 1$

dat.

$y_1 = y_2$

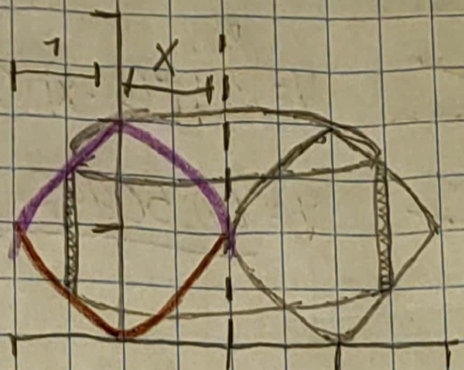
$x^2 = 2 - x^2$

$2x^2 = 2$

$x^2 = 1$

$x = -1, y = 1$

$x = 1, y = 1$



$r = 1 + x$

$h = y - y$

$h = 2 - x^2 - x^2$

$h = 2 - 2x^2$

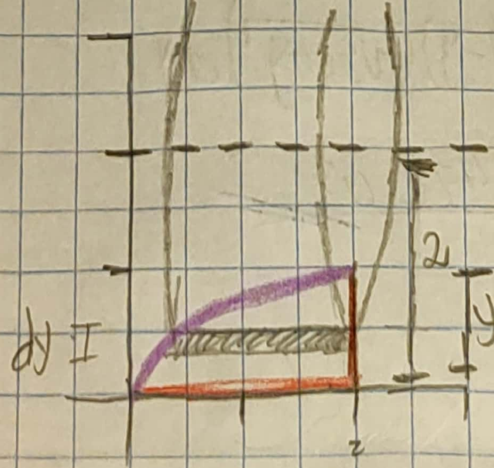
$V = 2\pi \int_{-1}^{1} (1+x)(2-2x^2)\,dx$

$V = 2\pi \int_{-1}^{1} 2 - 2x^2 + 2x - 2x^3\,dx$

$V = 2\pi \left(2x - \frac{2}{3}x^3 + x^2 - \frac{1}{2}x\right.$

$V = \frac{11}{3}\pi - \left(-\frac{5}{3}\pi\right) = \frac{16}{3}\pi$

19) $x = 2y^2$, $y \geq 0$, $x = 2$ alrededor $y = 2$



$r = 2 - y \quad h = 2 - 2y^2$

$V = 2\pi \int_0^1 (2-y)(2-2y^2)\,dy$

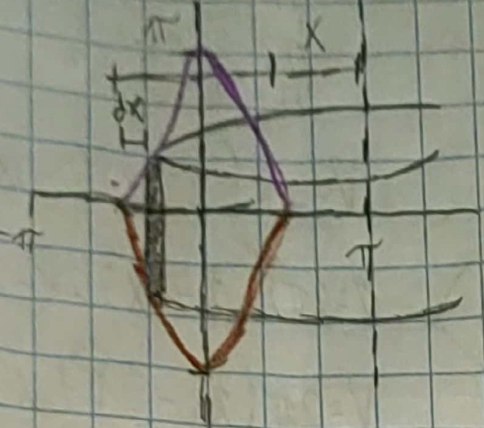
$V = 2\pi \int_0^1 4 - 4y^2 - 2y + 2y^3\,dy$

$V = 2\pi \left(4y - \frac{4}{3}y^3 - y^2 + \frac{1}{2}y^4\right)\Big|_0^1$

$V = \frac{13}{3}\pi\ u^3$

486 7309

23) $y = \cos^4 x$, $y = -\cos^4 x$, $-\pi/2 \le x \le \pi/2$ alrededor $x = \pi$



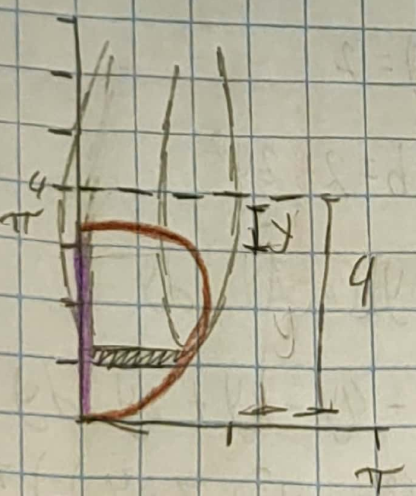
$r = \pi - x$ $h = y - y = \cos^4 x - (-\cos^4 x)$

$h = \cos^4 x + \cos^4 x = 2\cos^4 x$

$V = \int_{-\pi/2}^{\pi/2} 2\pi(\pi - x)(2\cos^4 x)\,dx$

$V = 46.50942\ u^3$

25) $x = \sqrt{\operatorname{sen} y}$, $0 \le y \le \pi$, $x = 0$ alrededor $y = 4$

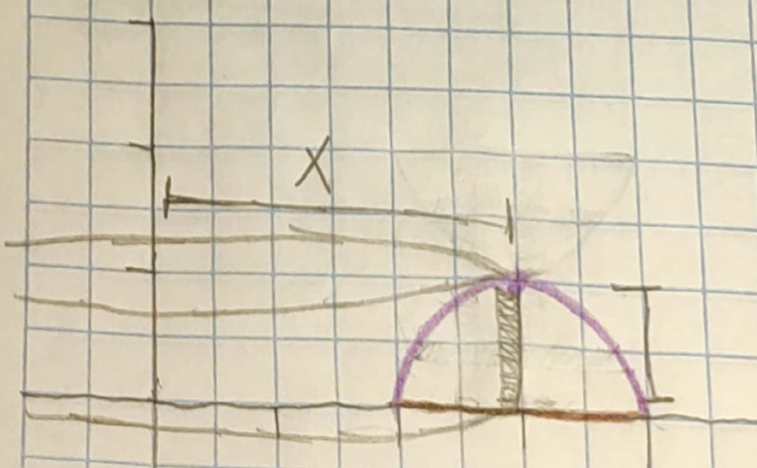


$r = 4 - y$ $h = \sqrt{\operatorname{sen} y}$

$V = 2\pi \int_0^{\pi} (4-y)(\sqrt{\operatorname{sen} y})\,dy$

$V = 36.57476\ u^3$

37) $y = -x^2 + 6x - 8$, $y = 0$ alrededor y



$r = x$ $h = y = -x^2 + 6x - 8$

$V = 2\pi \int_2^4 x(-x^2 + 6x - 8)\,dx$

$V = 2\pi \int_2^4 -x^3 + 6x^2 - 8x\,dx$

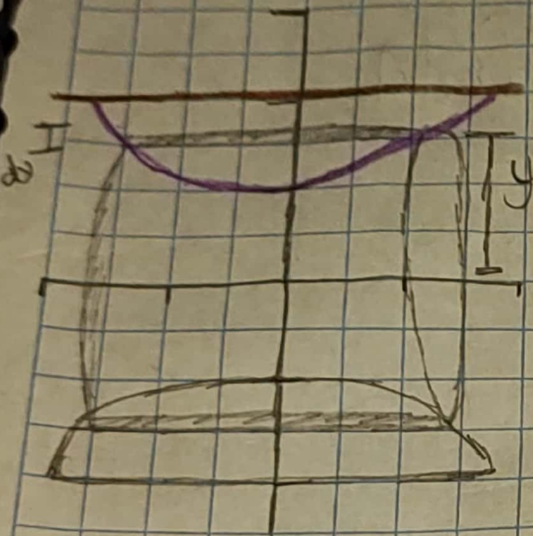
$V = 2\pi\left(-\frac{1}{4}x^4 + 2x^3 - 4x^2\right)\Big|_2^4$

$V = 0 - (-8\pi) = -8\pi$

39) $y^2 - x^2 = 1$, $y = 2$ alrededor x

$r = y$

$h = \sqrt{-1+y^2}$



$y^2 - x^2 = 1$

$-x^2 = 1 - y^2$

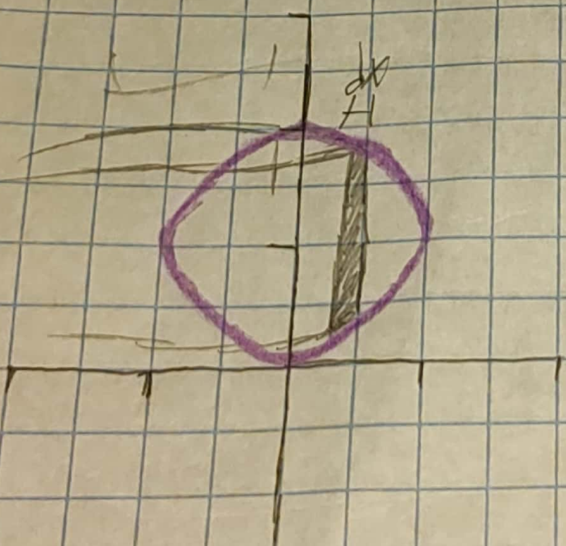
$x = \sqrt{-1+y^2}$

$V = 2\pi \int_1^2 y\left(\sqrt{-1+y^2}\right)$

$V = \sqrt{3}\,u^3$

Duda

41) $x^2 + (y-1)^2 = 1$ alrededor y $r = x$ $h = \sqrt{1-x^2}+1$



$V = 2\pi \int_{-1}^{1} x\left(\sqrt{1-x^2}+1\right)$

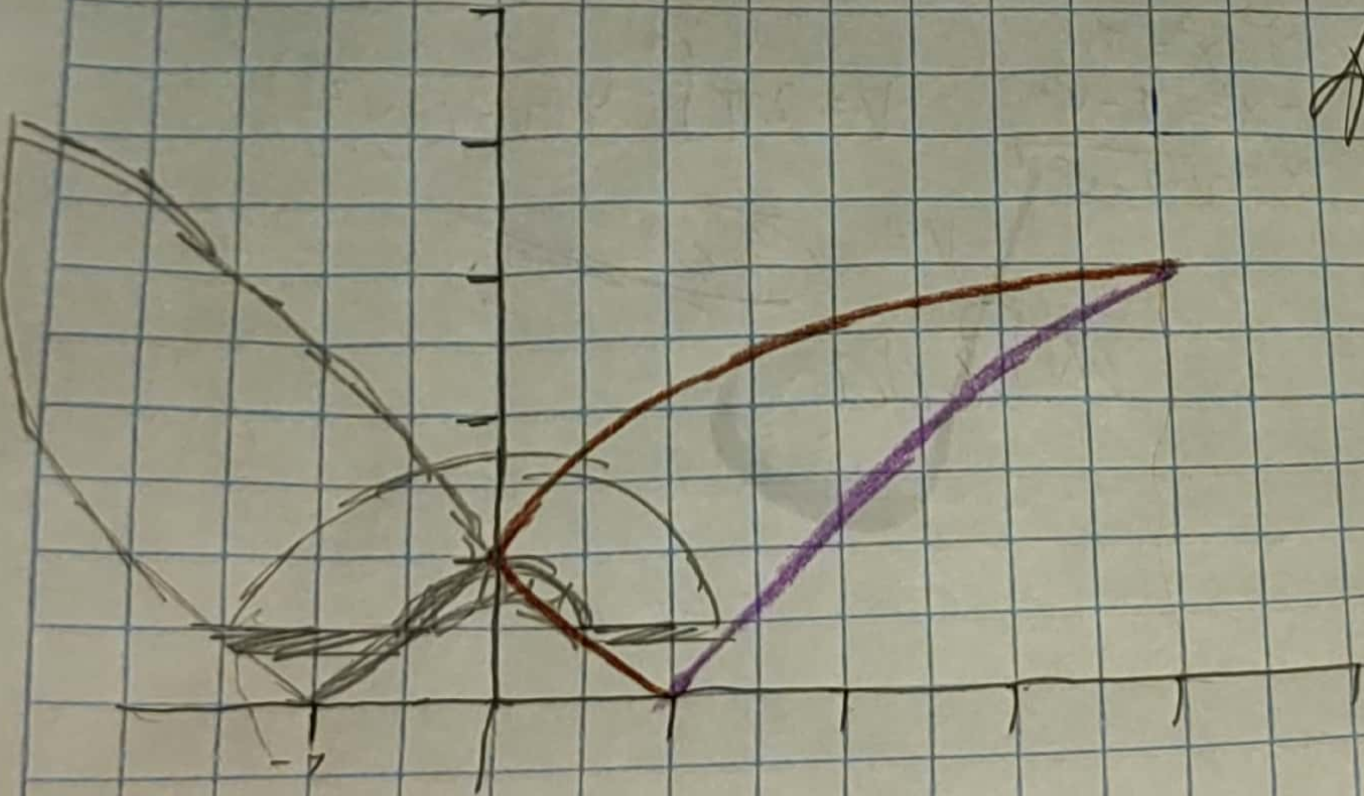
$V = 0$

Duda

$x^2 + (y-1)^2 = 1$

$y = \sqrt{1-x^2}+1$

43) $x = (y-1)^2, \quad x - y = 1$ alrededor $x = -1$



A Dada

$x = (y-1)^2$

$x = y + 1$